

Conversion of Internal Coordinates to Cartesian Coordinates

FORMULAE

$$\begin{aligned}
 \text{Atom 1} \quad p_1 &= (0, 0, 0) \\
 \text{Atom 2} \quad p_2 &= (r_{12}, 0, 0) \\
 \text{Atom 3} \quad p_3 &= (r_{12} - r_{23} \cos \theta_{123}, r_{23} \sin \theta_{123}, 0) \\
 \text{Atom 4} \quad e_1 &= (p_3 - p_2) / |p_3 - p_2| \\
 e_3 &= ((p_2 - p_1) \times e_1) / |(p_2 - p_1) \times e_1| \\
 e_2 &= e_3 \times e_1 \\
 \mathbf{M} &= [e_1 \quad e_2 \quad e_3]
 \end{aligned}$$

$$p_4 = p_3 + \mathbf{M} \cdot \begin{bmatrix} -r_{34} \cos \theta_{234} \\ r_{34} \sin \theta_{234} \cos \varphi_{1234} \\ r_{34} \sin \theta_{234} \sin \varphi_{1234} \end{bmatrix}$$

VARIABLES

SYMBOL	MEANING	UNITS
p_1, p_2, p_3, p_4	Cartesian position vectors of atoms 1 to 4. Atom 1 has no internal coordinates given, so it is taken as the origin, and the molecule's overall position and orientation are arbitrary.	Å
r_{12}	Bond length between atoms 1 and 2.	Å
r_{23}	Bond length between atoms 2 and 3.	Å
r_{34}	Bond length between atoms 3 and 4.	Å
θ_{123}	Bond angle at atom 2, between the bonds 2–1 and 2–3. It is the interior angle at atom 2, which is why $\cos \theta_{123}$ enters with a minus sign.	degrees
θ_{234}	Bond angle at atom 3, between the bonds 3–2 and 3–4.	degrees
φ_{1234}	Torsion (dihedral) angle about the 2–3 bond: the angle between the plane through atoms 1, 2, 3 and the plane through atoms 2, 3, 4. $\varphi = 0$ is the eclipsed (<i>cis</i>) arrangement; positive values rotate clockwise when looking from atom 2 toward atom 3.	degrees

e_1	Unit vector along the 2→3 bond, i.e. the bond being extended. Local x axis.	—
e_3	Unit vector normal to the plane through atoms 1, 2, 3. This is the axis that φ_{1234} rotates about. Local z axis.	—
e_2	Unit vector completing the right-handed set, $e_3 \times e_1$. Local y axis.	—
M	3×3 matrix holding e_1, e_2, e_3 as its <i>columns</i> . Because $e_1 \times e_2 = e_3$ it is a pure rotation, so it carries the bracketed vector from the local frame into world coordinates without changing its length.	—
$\times$	Vector cross product.	—
$ v $	Euclidean length of the vector v , used to normalise to unit length.	—
[...]	The position of atom 4 relative to atom 3, written in spherical polar form in the local $e_1 e_2 e_3$ frame. Its length is r_{34} .	Å